SOFTWARE REQUIREMENTS SPECIFICATION

ARYA COLLEGE CLUBSPHERE

College Club and Event Management System

Prepared for
Arya College of Engineering & IT, Jaipur

Prepared by
Arya College ClubSphere Team

Technology Stack
Java • JSP • Servlets • JDBC • MySQL • Apache Tomcat

Version 1.0 | 28 October 2026

This document describes the requirements represented in the project README. Details that depend on the current implementation are identified as such.

Document Control

Item	Details
Document Title	Software Requirements Specification (SRS)
Project Name	Arya College ClubSphere
Project Type	Web-based College Club and Event Management System
Institution	Arya College of Engineering & IT, Jaipur
Version	1.0
Date	28 October 2026
Primary Source	Project README supplied for this document
Status	Requirements baseline for academic project documentation; verify against the final integrated codebase.

Table of Contents

- 1. Introduction
- 2. Overall Description
- 3. System Features and Functional Requirements
- 4. External Interface Requirements
- 5. Non-Functional Requirements
- 6. Data and Database Requirements
- 7. Use Cases
- 8. System Environment and Constraints
- 9. Assumptions and Dependencies
- 10. Future Enhancements
- 11. Team Module Allocation
- 12. Acceptance Criteria
- 13. Conclusion

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the intended capabilities, user roles, interfaces, constraints, and quality requirements of Arya College ClubSphere. It is intended to guide development, integration, testing, and academic review of the system.

1.2 Scope

Arya College ClubSphere is a web-based platform developed for Arya College of Engineering & IT, Jaipur. Its goal is to centralize and simplify college club activities, student memberships, event information, event registrations, and related tasks. The system is intended to reduce manual coordination and make relevant club and event information easier for users to access.

1.3 Intended Audience

• Project team members responsible for implementation and integration.

• Faculty members, project guides, and evaluators reviewing the system.

• Administrators and club heads who manage club and event activities.

• Students and club members who explore and participate in campus activities.

1.4 Definitions and Acronyms

Term	Meaning
SRS	Software Requirements Specification
JSP	JavaServer Pages; used to render dynamic web pages
Servlet	Java component that handles web requests and responses
JDBC	Java Database Connectivity
DBMS	Database Management System
QR Code	Quick Response code used by the described event-registration/check-in workflow
Role-based access	Permission to use features based on the user's assigned role

2. Overall Description

2.1 Product Perspective

ClubSphere is a browser-based application using a Java web application architecture. The README identifies HTML, CSS, JavaScript, and JSP for the web interface; Java Servlets for request handling and application logic; JDBC for database connectivity; MySQL for persistent data; and Apache Tomcat as the application server.

2.2 Product Functions

• Authenticate users through available registration and login functionality.

• Apply role-based access to permitted features.

• Maintain information about college clubs and memberships.

• Manage college event information and student registrations.

• Support QR-based event registration/check-in where implemented.

• Provide role-appropriate dashboards and support task-related activities and notifications.

• Store and retrieve relevant application information through the database.

2.3 User Classes and Characteristics

User Role	Description / Responsibilities
Admin	Manages the overall system and performs authorized administrative activities.
Club Head	Manages assigned club activities, coordinates events, and handles relevant event registrations.
Club Member	Participates in club activities and accesses permitted club-related features.
Student	Explores available clubs and events and participates in eligible college activities.

2.4 Operating Environment

• Client: a modern web browser on a computer or compatible device.

• Development/runtime environment identified in the README: JDK 17, Apache Maven 3.9 or later, Apache Tomcat 10.1, and MySQL Server 8.0 or later.

• Development tools: Visual Studio Code or another compatible Java IDE, Git, and GitHub.

2.5 Design and Implementation Constraints

• The implementation uses Java, JSP, Servlets, JDBC, MySQL, Maven, and Apache Tomcat as described in the README.

• Database configuration, schema, table names, and SQL queries must remain consistent.

• Access to administrative and club-management actions must be restricted according to roles.

• The deployed context path and actual runtime behavior depend on the final build and Tomcat configuration.

3. System Features and Functional Requirements

The requirements below are derived from the features and workflow described in the supplied README. Where the README does not specify a detailed behavior, the requirement is stated at a high level and should be verified against the implemented application.

ID	Feature	Requirement
FR-01	Authentication and User Management	**1.** The system shall provide the available user registration and login functionality. **2.** The system shall identify a user's role after successful authentication and restrict features according to assigned permissions.
FR-02	Club Management	**1.** The system shall allow authorized users to manage club information and related details. **2.** The system shall support club-membership information where the corresponding module is implemented.
FR-03	Event Management	**1.** The system shall support the management of college event information. **2.** Authorized organizers shall be able to manage relevant event registrations according to their permissions.
FR-04	Event Registration	**1.** The system shall provide the available process for students to register for eligible events. **2.** The system shall retain registration information in the database when supported by the implementation.
FR-05	QR-based Registration / Check-in	**1.** The system shall support the QR-based event-registration or check-in workflow described in the README. **2.** QR generation, scanning, and verification details shall follow the actual implemented workflow; the README does not define every technical step.
FR-06	Role-based Dashboards	**1.** The system shall provide dashboard functionality appropriate to a user's role and permitted activities. **2.** A user shall not be granted access to restricted dashboard features merely by changing a URL or request.

ID	Feature	Requirement
FR-07	Tasks and Notifications	**1.** The system shall support task-related activities and notifications where available in the integrated application. **2.** The exact notification channels and triggers depend on the implemented module.
FR-08	Database Operations	**1.** The system shall store and manage application data related to users, clubs, events, registrations, and other implemented modules. **2.** The application shall use JDBC to communicate with the configured MySQL database.
FR-09	Request Processing	**1.** The application shall process browser requests through the Java web application components and return an appropriate response. **2.** Errors should be handled without exposing sensitive internal information.

3.1 Authorization Rules

• The system should associate each authenticated account with an assigned role.

• Only authorized users should be able to perform administrative operations.

• Club heads should be limited to the club activities and event operations assigned to them.

• Club members and students should access only the features permitted to their roles.

• The server side should validate permissions for protected operations.

4. External Interface Requirements

4.1 User Interface

Users interact with the system through web pages rendered using the frontend technologies identified in the README. The interface should make the available actions understandable for each role and provide clear feedback for successful actions, validation errors, and failures.

4.2 Software Interfaces

Interface	Requirement
Web Browser	Users access the application through a supported browser using the deployed application URL.
Java Servlets and JSP	Servlets process requests and JSP pages provide dynamic web content, as implemented.
JDBC and MySQL	The application uses JDBC to connect to the configured MySQL database.
Apache Tomcat	The web application is deployed and executed using the configured Tomcat environment.
Git and GitHub	The team uses version control to coordinate development and integration.

4.3 Communication Interface

The application is accessed through a browser using the URL and context path configured for the deployment. The README provides an example local Tomcat URL; the actual URL may differ depending on the deployed WAR filename and server configuration.

4.4 Hardware Interface

No specialized hardware interface is specified in the README. A computer or compatible device capable of running a web browser is sufficient for normal user interaction. QR scanning/check-in may use a device camera or another QR-reading method if supported by the implemented workflow.

5. Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR-01	Security	Passwords must be handled using secure password practices. Database credentials must not be committed to a public repository. Protected actions must enforce role permissions.
NFR-02	Input Validation	User-provided data should be validated before processing, including on the server side for important operations.
NFR-03	Database Safety	Parameterized SQL queries or PreparedStatement should be used to reduce SQL injection risks.
NFR-04	Reliability	The application should handle invalid input, unavailable database connections, and unexpected errors gracefully.
NFR-05	Maintainability	The project should retain a clear module structure and use version control to coordinate changes.
NFR-06	Compatibility	The application should be compatible with the Java, Servlet API, Maven, Tomcat, and MySQL versions selected for the project.
NFR-07	Usability	The user interface should present role-relevant actions and clear feedback for common operations.
NFR-08	Data Integrity	User, club, event, and registration records should be stored and updated consistently according to the database schema.
NFR-09	Performance	Common user actions should complete in a reasonable time under the expected college-project workload. A numeric response-time target is not specified in the README.

6. Data and Database Requirements

6.1 Database Platform

The README identifies MySQL 8 as the database platform and JDBC as the database connectivity layer. The database name, table names, relationships, and SQL schema must match the actual application configuration and queries.

6.2 Data Categories

• User accounts and role information.

• College club details and membership information.

• Event details and student registration records.

• QR-related registration or check-in records, where implemented.

• Task and notification information, where implemented.

6.3 Data Handling Requirements

• The application should retrieve and update records through the configured database connection.

• Database credentials should be kept private and should not be stored in publicly committed configuration.

• Validation should be performed before saving important data.

• The database schema should be maintained alongside the project so that a new development environment can be configured consistently.

Note: The supplied README describes possible data categories but does not provide the complete database schema or exact table definitions. Those details should be taken from the project's SQL scripts and source code.

7. Use Cases

Use Case	Name	Primary Actor	Main Flow
UC-01	Register / Log in	Student, Club Member, Club Head, Admin	User submits the available registration or login information. The application processes the request and, on successful authentication, provides access according to the assigned role.
UC-02	Explore Clubs	Student	Student views available club information and uses the club-related features made available to the student role.
UC-03	Manage Club Information	Admin / authorized Club Head	Authorized user accesses the club-management module and performs permitted club-related operations.
UC-04	Manage an Event	Admin / authorized Club Head	Authorized organizer accesses event management and creates or updates event information according to the implemented functionality.
UC-05	Register for an Event	Student / eligible user	User follows the available registration process for an event. Registration data is recorded when the module and database operation are implemented.
UC-06	Verify Event Check-in	Authorized organizer	Organizer uses the QR-based check-in workflow, where implemented, to help verify event registration.
UC-07	Access Dashboard	All authenticated roles	User opens the role-appropriate dashboard and accesses only permitted functions.
UC-08	Use Tasks / Notifications	Permitted users	User views or interacts with task-related activities and notifications provided by the integrated module.

7.1 General Exception Scenarios

• If required input is missing or invalid, the system should display an appropriate message and avoid processing invalid data.

• If authentication fails, access to protected functionality should not be granted.

• If a user attempts an unauthorized operation, the system should deny the operation.

• If a database operation fails, the application should handle the error safely and avoid displaying private configuration or sensitive technical details.

8. System Environment and Constraints

8.1 Software Requirements

Software / Tool	Version / Purpose
Java Development Kit	JDK 17
Apache Maven	3.9 or later, as stated in the README
Apache Tomcat	10.1
MySQL Server	8.0 or later
IDE	Visual Studio Code or another compatible Java IDE
Version Control	Git and GitHub
Browser	A modern browser capable of accessing the deployed web application

8.2 Build and Deployment

The README describes a Maven-based Java web application. The project is built from the directory containing pom.xml. If configured for WAR packaging, Maven generates a WAR artifact under the target directory, which is deployed to the configured Apache Tomcat instance.

8.3 Security Constraints

• Real database passwords and private credentials must not be included in source control.

• The application should use server-side authorization for protected actions.

• Database operations should use safe query practices and validate incoming data.

9. Assumptions and Dependencies

• Users have access to a compatible web browser and a working network connection to the application.

• The MySQL service is running and the configured database and required tables are available.

• The Java, Maven, Tomcat, and MySQL versions are configured compatibly with the application's dependencies.

• Role assignments and permissions are configured by the application according to its implemented authentication and authorization logic.

• The exact behavior of QR generation, scanning, verification, tasks, and notifications depends on the final integrated codebase; the README does not fully specify every implementation detail.

• The project is developed collaboratively by four team members and integrated using Git and GitHub.

10. Future Enhancements

The README lists the following as potential future enhancements. They should not be treated as implemented features unless they are present in the current codebase.

• Improved role-based dashboards.

• Enhanced club membership management.

• Advanced event scheduling and management.

• Improvements to QR-based event check-in.

• Real-time notifications and email notifications for important events.

• Event attendance reports and club activity reports.

• Improved mobile responsiveness.

• Additional validation and automated tests.

• Deployment to a hosted environment.

11. Team Module Allocation

S. No.	Team Member	Enrollment Number	Assigned Module
1	Avantika Mishra	24EARCS032	Authentication & User Management
2	Avanshi Kothari	24EARCS031	Club Management
3	Akansha Sharma	24EARCS012	Event Management & QR Code Registration
4	Aditi Maheshwari	24EARCS007	Dashboards, Tasks & Notifications

Each team member is responsible for the assigned module and coordinates with the other members to integrate the application. GitHub is used for source-code management and collaboration, according to the README.

12. Acceptance Criteria

The following high-level criteria can be used during project review. Final verification should be performed against the features actually implemented in the integrated application.

• The application builds successfully using the project's configured Maven setup.

• The application can be deployed to the configured Tomcat environment.

• The application can connect to the configured MySQL database.

• Available registration and login functionality behaves as intended.

• Users receive access according to their assigned roles and permissions.

• Implemented club and event features can be accessed by authorized users.

• The event-registration and QR-based check-in workflow can be demonstrated if it is implemented in the integrated build.

• Implemented dashboards, tasks, and notifications behave consistently with their available functionality.

• Invalid input and unauthorized requests are handled safely.

• No real database passwords or other private credentials are exposed in the repository.

13. Conclusion

Arya College ClubSphere is intended to provide a centralized web-based system for college clubs, memberships, events, and event registrations. This SRS records the system's high-level purpose, user roles, functional requirements, interfaces, quality requirements, data needs, constraints, and team responsibilities based on the supplied README. Before final submission, the project team should compare this document with the final source code and database schema so that every described requirement accurately reflects the integrated implementation.

End of Software Requirements Specification